

HSIL Hackathon – Concept Note

NeuroSense: Passive Smartphone Monitoring for Parkinson's Disease

A Family-Centred Remote Monitoring Platform | Team Beirut 332

Work in Progress: This concept note reflects our thinking at the start of the hackathon. As we progress, findings from research, testing, user feedback, and new data or technologies may require us to refine or adjust aspects of the proposed solution. We welcome mentor input as part of this iterative process.

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Problem Statement

Parkinson's disease (PD) affects 10 million people worldwide and grows fastest in low- and middle-income countries (LMICs). Its core medication — levodopa — has a ~90-minute half-life; a delay of just 30–60 minutes triggers severe symptom flare-ups. Yet patients see a neurologist only every 3–6 months, leaving clinicians to titrate doses based on brief, subjective in-clinic snapshots.

This gap hits hardest for three groups:

- **Patients:** Elderly and advanced-stage individuals who cannot sustain frequent specialist visits due to financial, logistical, or physical constraints.
- **Families and caregivers:** PD is a whole-family burden. Caregivers monitor daily fluctuations and manage medications with no structured support or objective data.
- **Neurologists:** In Lebanon and across MENA, roughly 1 movement disorders specialist exists per million people, making remote monitoring essential.

Existing digital tools (StrivePD, Roche PD App, PKG Watch) require Apple Watches and high digital literacy, structurally excluding the patients who need monitoring most. Phone-based passive monitoring is underexplored, and the family/caregiver-centred design axis is absent from all FDA-approved devices in this space.

Proposed Solution

NeuroSense is an Android app that passively monitors PD symptoms using only the phone's built-in IMU — no smartwatch, no wearable, no daily patient engagement required. It is operated primarily by a family caregiver or visiting nurse, and delivers tiered reports to both the family and the treating neurologist.

- **Passive monitoring (primary):** Phone carried in pocket by day, optionally placed under the pillow at night. Background services sample the accelerometer and gyroscope at 50 Hz, extracting biomarkers: tremor spectral power, gait cadence, step regularity, freezing-of-gait events, sit-to-stand frequency, and nocturnal movement patterns.
- **Active tests (optional, caregiver-facilitated):** Brief tasks triggered when a caregiver or nurse is present: sustained tremor hold, timed finger-tap, short walking pass. Augment passive data with MDS-UPDRS-aligned assessments.
- **AI Layer 1 — ON/OFF classifier:** An ML model trained on public PD accelerometry datasets (Levodopa Response Study, mPower) classifies day windows as medication ON or OFF, producing a wearing-off timeline that directly informs titration decisions.
- **AI Layer 2 — LLM report generation:** De-identified trend data is sent to an LLM API (Gemini or open-source), generating a concise weekly clinical narrative with flagged events and suggested titration discussion points — readable in under two minutes. Patient identity is redacted before leaving the device.
- **Three user roles:** Patient (passive); Caregiver/nurse (setup, optional tests, wellness summary); Neurologist (clinical report).

Hackathon scope: one phone, public PD dataset or simulated IMU data, pre-trained classifier, LLM-generated report. No proprietary hardware or patient recruitment required.

Special / Innovative Features

- **Passive-first, caregiver-operated:** Serves advanced-stage patients that all incumbents structurally exclude — those who cannot wear a smartwatch or engage daily with an app.
- **Wearing-off timeline:** The ON/OFF classifier directly answers the clinical question driving titration. No consumer tool currently provides this from phone IMU alone.
- **Family framing:** Tiered reports — clinical narrative for the neurologist, wellness summary for the caregiver — a design axis absent from all current FDA-approved PD devices.
- **Android-first, LMIC-ready:** Runs on sub-\$100 phones with offline buffering and multilingual UI, targeting geographies where specialist access is worst.
- **Dual AI architecture:** ML classifier for time-series pattern recognition + LLM for clinical synthesis. Together they cover technical depth and clinical usability.
- **Privacy by design:** Only de-identified numerical trends leave the device. No video, no images, no identifiable data on external servers.

Expected Benefits / Impact

Short-term

- Objective between-visit data enabling evidence-based medication titration.
- Earlier detection of wearing-off before falls or ER visits.
- Reduced caregiver burden through passive, structured monitoring.
- Access extended to patients excluded from all existing tools.

Long-term

- **System shift:** From episodic clinic snapshots to continuous objective monitoring — as in cardiology and diabetes care.
- **Cost reduction:** Fewer preventable hospitalisations and delayed nursing-home placement.
- **Equity:** Specialist capacity extended to rural and LMIC patients via async remote monitoring.
- **Expandable platform:** Sensor and AI infrastructure transfers directly to ALS, Huntington's, and atypical parkinsonisms.

Feasibility & Implementation

Roadmap

- **Phase 1 — Hackathon:** Working demo: phone + public dataset + ON/OFF classifier + LLM report.
- **Phase 2 — Months 1–3:** Clinical validation, 20–30 patients at one neurology clinic (e.g. AUB), app features vs. MDS-UPDRS. IRB required.
- **Phase 3 — Months 3–12:** LMIC pilot via HSIL hub partner, 200+ patients, peer-reviewed publication.
- **Phase 4 — Year 2:** SaMD regulatory pathway, pharma real-world evidence partnerships, RPM reimbursement.

Resources

- **Hackathon:** Android dev environment, LLM API credits, public PD accelerometry dataset.
- **Post-hackathon:** IRB approval, clinic partnership, ~\$50K for six-month validation study.

Risks & Mitigations

- **Clinical validation:** Early specialist partnership; use of literature-validated biomarker targets.
- **IMU accuracy vs. wearables:** Focus on robust population-level patterns; flag data quality in reports.
- **Regulatory timeline:** Clinical decision-support framing defers SaMD classification to Phase 4.
- **Caregiver adherence:** Passive mode requires no daily action; active tests are optional and brief.
- **Competition:** Incumbents cannot easily pivot to serve this population without redesigning their core architecture.